

All about me

- BORN: 1791
- DIED: 1867
- NATIONALITY: English
- FACTOID: My face has been on the £20 note.
- IN A NUTSHELL: I was born near London, England. My father was a blacksmith, and I had very little formal education.

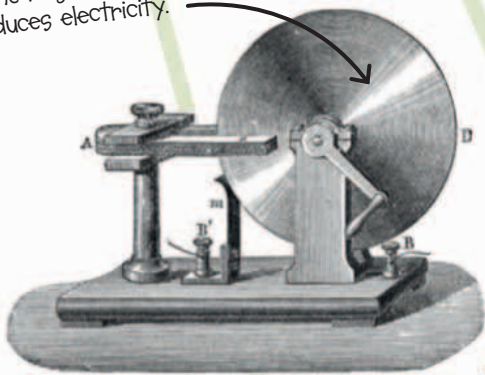
Michael Faraday

The man who invented the electric MOTOR and the electric generator

A notion of motion

In 1821, Faraday showed that when you flow electricity through a coil of wire, it creates a **magnetic field**. He realized that this electromagnetic energy could be used to produce **motion**, and invented the **ELECTRIC MOTOR**, which is still used in countless ways today.

When a copper disk is spun past the poles of the magnet, it produces electricity.



All in a spin

Faraday continued to experiment with magnetism, and, in 1831, he realized that if he **reversed his process** he could produce an electrical current. By spinning a copper disk between the poles of a magnet, he could **generate a steady flow of electricity** through a wire. Faraday had invented the first **DYNAMO** (pictured above), which would eventually become the electric generator.

By the way...

I discovered that a metal "Faraday" cage can protect whatever is inside from a huge electric current, by conducting it safely around the cage.

Faraday designed the first transformer.



These are now used to turn high voltages into low ones that are safe enough to use in our homes.

